

## Homework (Jalapeno)

- Q1.** In ancient Babylon, a tablet was discovered with the equation  $x = 2 \cdot \sqrt{3} + 3 \cdot \sqrt{3}$ . Simplify the equation.  
(A)  $5 \cdot \sqrt{3}$   
(B)  $2 \cdot \sqrt{3} + 3 \cdot \sqrt{2}$   
(C)  $\sqrt{3} + \sqrt{2}$   
(D)  $\sqrt{5}$   
(E)  $\sqrt{2}$
- Q2.** The Great Pyramid of Giza has a base perimeter of  $2 \cdot \sqrt{2} + 2 \cdot \sqrt{3}$  kilometers. Simplify the expression.  
(A)  $2 \cdot (\sqrt{2} + \sqrt{3})$   
(B)  $\sqrt{2} + \sqrt{3}$   
(C)  $2 \cdot \sqrt{2}$   
(D)  $2 \cdot \sqrt{3}$   
(E)  $\sqrt{2} \cdot \sqrt{3}$
- Q3.** The Rosetta Stone has an inscription with the equation  $\sqrt{2} \cdot \sqrt{6} =$   
(A)  $\sqrt{12}$   
(B)  $2 \cdot \sqrt{3}$   
(C)  $\sqrt{3}$   
(D)  $\sqrt{2}$   
(E)  $\sqrt{6}$
- Q4.** A Roman aqueduct has a length of  $3 \cdot \sqrt{2}$  meters and a width of  $2 \cdot \sqrt{2}$  meters. Find the area.  
(A)  $6 \cdot \sqrt{2}$   
(B) 6  
(C) 12  
(D)  $12 \cdot \sqrt{2}$   
(E)  $\sqrt{2}$
- Q5.** The ancient Greeks used the equation  $\frac{\sqrt{3}}{\sqrt{2}} =$   
(A)  $\frac{\sqrt{6}}{2}$   
(B)  $\frac{\sqrt{2}}{\sqrt{3}}$   
(C)  $\sqrt{2} \cdot \sqrt{3}$   
(D)  $\frac{\sqrt{2}}{2}$   
(E)  $\frac{\sqrt{3}}{2}$
- Q6.** The Hanging Gardens of Babylon had a height of  $2 \cdot \sqrt{3}$  meters and a width of  $3 \cdot \sqrt{3}$  meters. Find the area.  
(A)  $6 \cdot \sqrt{3}$   
(B) 6  
(C) 12  
(D)  $12 \cdot \sqrt{3}$   
(E)  $\sqrt{3}$
- Q7.** The ancient Egyptians used the equation  $\sqrt{2} \cdot \sqrt{8} =$   
(A) 4  
(B)  $\sqrt{16}$   
(C)  $\sqrt{4}$   
(D)  $2 \cdot \sqrt{2}$   
(E)  $\sqrt{2}$
- Q8.** The ancient Chinese used the equation  $\frac{\sqrt{6}}{\sqrt{2}} =$   
(A)  $\sqrt{3}$   
(B)  $\sqrt{2}$   
(C)  $\frac{\sqrt{6}}{2}$   
(D)  $\frac{\sqrt{2}}{\sqrt{3}}$   
(E)  $\frac{\sqrt{3}}{2}$

### Open Ended Questions (FRQ)

- Q9.** The ancient Greeks built a temple with a length of  $4 \cdot \sqrt{2}$  meters and a width of  $3 \cdot \sqrt{2}$  meters. Find the area and perimeter of the temple.
- Q10.** A Roman engineer designed a bridge with a length of  $5 \cdot \sqrt{3}$  meters and a width of  $2 \cdot \sqrt{3}$  meters. Find the area and perimeter of the bridge.

### Chef's Tips

- Tip 1: Encourage students to use historical contexts to understand the relevance of radicals.
- Tip 2: Emphasize the importance of simplifying radical expressions.
- Tip 3: Remind students to check their calculations carefully.